FINARTS C01

PROGRESS REPORT

GROUP 4

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CHOSEN ARTICLE & Methodology

Beirne, J. & Sugandi, E. (2023). Risk-off shocks and spillovers in safe havens. Pacific-Basin Finance Journal, 80, 102102.

Model: Country-Specific Structural Vector Autoregression (SVAR)

Same model per country

Identification Strategy: Cholesky Decomposition

Orders variables from most exogenous (external global shocks) to most endogenous (domestic capital flows).

Variable Ordering

1. Global Shocks: Risk-off binary, Log(WUI)
2. Financial Markets: Yield Spread, Log(Nikkei 225)
3. Macroeconomy: Log(RGDP), Log(REER)
4. Capital Flows (% GDP): Debtsec, Equity, Other, Direct

Hypothesis from the Paper

- H1: Global risk-off shocks lead to a sharp appreciation of the Japanese Yen.
- H2: Risk-off shocks do not trigger significant physical portfolio capital inflows into Japan.
- H3: Global risk-off shocks cause negative spillovers to Japan's real economy (GDP contraction).

Replication Modifications - Sample & Scope Replication Modifications

- Scope Shift:
  - Focusing exclusively on Japan (removing Switzerland and the US comparisons)
    - To isolate and retest the "Safe Haven Yen" theory.

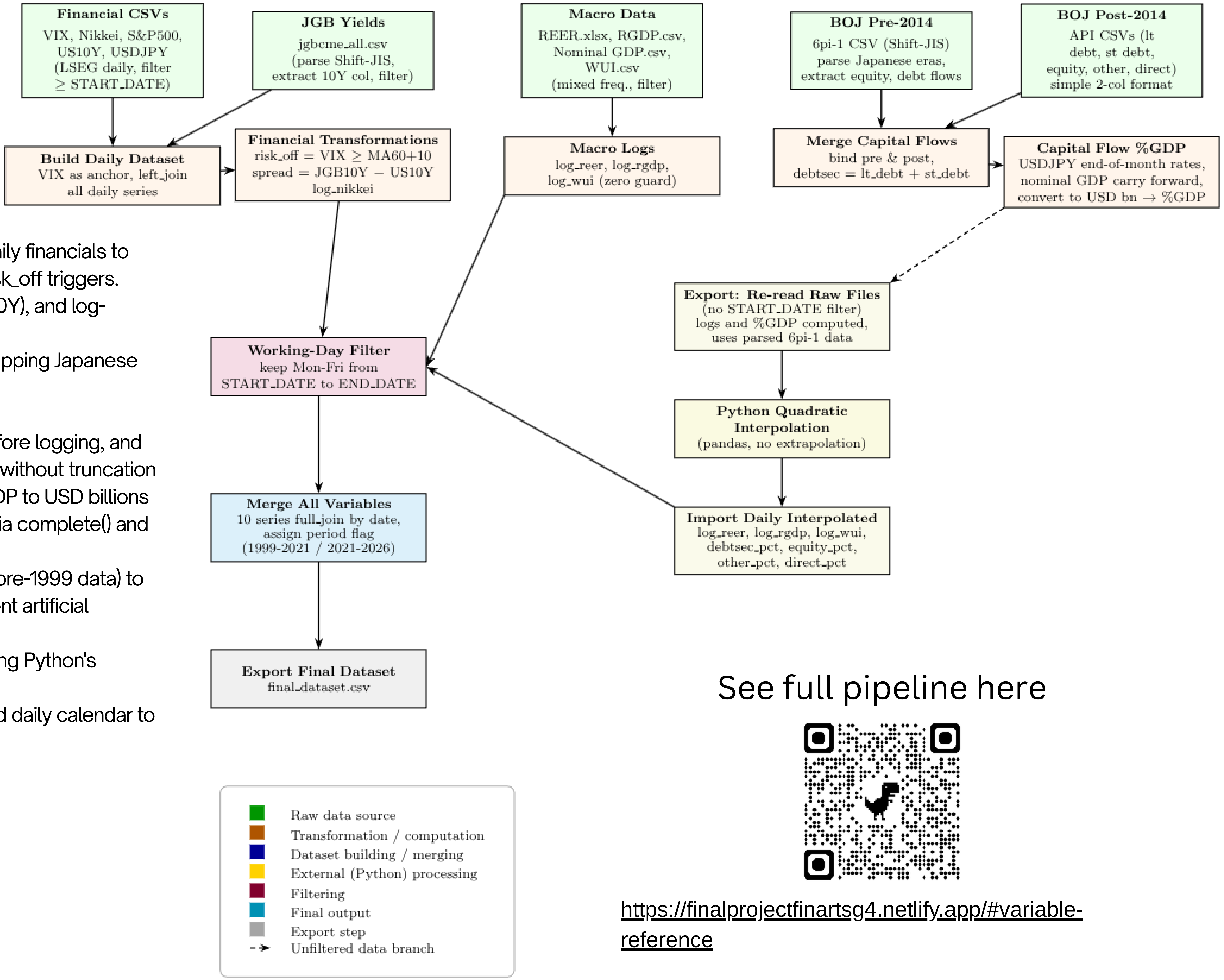
- Extended the original sample (1999–2021) up to June 30, 2026
  - This extension captures recent macroeconomic anomalies, including:
    - Historic Yen depreciation
    - BOJ policy normalization and rate hikes
    - Recent risk-off episodes (e.g., Iran conflict, carry trade unwind)
    - "America First" Trade Policies
    - BOJ Normalization vs. Global Pressures
    - Record Currency Interventions
- Data
  - Higher Frequency Data:
    - Replaced the original paper's **quarterly** IMF/CEIC sourced capital flow data with **monthly** Bank of Japan (BOJ) sourced data.
  - Direct Central Bank Sourcing & LSEG / Bloomberg

Variables, Data, & Data Source

Variable	Description	Freq.	Source
`risk_off`	Risk-off indicator: 1 when VIX is at least 10 points above its 60-day MA	Daily	Bloomberg (VIX); Authors' calculation
`spread`	Yield spread between 10-year JGB and 10-year US Treasury	Daily	Japan MoF, LSEG
`log_nikkei`	Log of Nikkei-225 stock index	Daily	LSEG
`log_reer`	Log of real effective exchange rate, broad index (2010 = 100)	Monthly $\rightarrow$ daily	BIS
`log_rgdp`	Log of constant-price real GDP	Quarterly $\rightarrow$ daily	IMF
`log_wui`	Log of World Uncertainty Index for Japan	Quarterly $\rightarrow$ daily	worlduncertaintyindex.com
`debtsec_pct`	Net portfolio debt securities inflows (% of nominal GDP)	Monthly $\rightarrow$ daily	BOJ, IMF, LSEG
`equity_pct`	Net portfolio equity inflows (% of nominal GDP)	Monthly $\rightarrow$ daily	BOJ, IMF, LSEG
`other_pct`	Net other investment inflows (% of nominal GDP)	Monthly $\rightarrow$ daily	BOJ, IMF, LSEG
`direct_pct`	Net direct investment inflows (% of nominal GDP)	Monthly $\rightarrow$ daily	BOJ, IMF, LSEG

DATA PIPELINE & FEATURE ENGINEERING

- Filtered raw series to 1999-01-14–2026-06-30, anchoring daily financials to VIX via left_join to preserve US trading days and accurate risk_off triggers.
- Computed risk_off (VIX ⩾ MA60+10), spread (JGB10Y - US10Y), and log-transformed Nikkei
- Shift-JIS encoded BOJ files (1999–2013) using readLines, mapping Japanese eras (Heisei/Reiwa) to Gregorian years
- Combined cut off datasets 1999-2014 + 2014-2026
- Converted "-" to NA_real_ in JGB yields, filtered zero WUI before logging, and used full_join to preserve variable-specific temporal ranges without truncation
- Converted 100M JPY BOJ flows and millions-JPY nominal GDP to USD billions using EOM USD/JPY, expanding quarterly GDP to monthly via complete() and forward-fill..
- Exported all native-frequency raw files unfiltered (retaining pre-1999 data) to supply boundary anchors for the quadratic spline and prevent artificial extrapolation.
- Transformed low-frequency variables to daily frequency using Python's pandas.interpolate(method='quadratic')
- Applied lubridate::wday() %in% 2:6 to restrict the interpolated daily calendar to Monday–Friday trading days.
- Merged ten endogenous variables via date-based full_join



See full pipeline here

https://finalprojectfinartsg4.netlify.app/#variable-reference

Descriptive Statistics by Period

Variable	Stat	1999-2021	2021-2026	Full Sample
debtsec_pct	N	5795	1369	7164
	Mean	0.5164	0.6029	0.5329
	SD	1.7154	2.6632	1.9328
	Min	-6.1373	-5.3554	-6.1373
	Max	6.7079	7.0383	7.0383
direct_pct	N	5795	1369	7164
	Mean	0.1085	0.2489	0.1353
	SD	0.2921	0.4233	0.3260
	Min	-1.2558	-1.3029	-1.3029
	Max	3.6918	1.8328	3.6918
equity_pct	N	5795	1369	7164
	Mean	0.2174	0.3155	0.2361
	SD	1.1626	1.3304	1.1970
	Min	-3.7484	-3.5553	-3.7484
	Max	4.7229	3.8500	4.7229
log_nikkei	N	5260	1235	6495
	Mean	9.5565	10.4582	9.7279
	SD	0.3263	0.2404	0.4716
	Min	8.8615	10.1153	8.8615
	Max	10.3244	11.1895	11.1895
log_reer	N	5795	1369	7164
	Mean	4.5157	4.0614	4.4289
	SD	0.1711	0.0930	0.2393
	Min	4.2050	3.9183	3.9183
	Max	4.8758	4.2659	4.8758
log_rgdp	N	5795	1369	7164
	Mean	13.1932	13.2802	13.2099
	SD	0.0507	0.0096	0.0571
	Min	13.0888	13.2582	13.0888
	Max	13.2810	13.2933	13.2933
log_wui	N	5795	1369	7164
	Mean	-1.7987	-1.9129	-1.8205
	SD	0.5928	1.0512	0.7052
	Min	-3.0317	-3.9128	-3.9128
	Max	-0.6186	0.1929	0.1929
other_pct	N	5795	1369	7164
	Mean	0.3918	1.2345	0.5529
	SD	3.1556	4.2007	3.3962
	Min	-10.6442	-7.7495	-10.6442
	Max	28.5474	15.4066	28.5474
risk_off	N	5530	1317	6847
	Mean	0.0318	0.0190	0.0294
	SD	0.1756	0.1365	0.1688
	Min	0.0000	0.0000	0.0000
	Max	1.0000	1.0000	1.0000
spread	N	5259	1234	6493
	Mean	-2.4367	-2.7014	-2.4870
	SD	0.9313	0.7739	0.9094
	Min	-5.0590	-4.1430	-5.0590
	Max	-0.4960	-1.1520	-0.4960

Descriptive Statistics

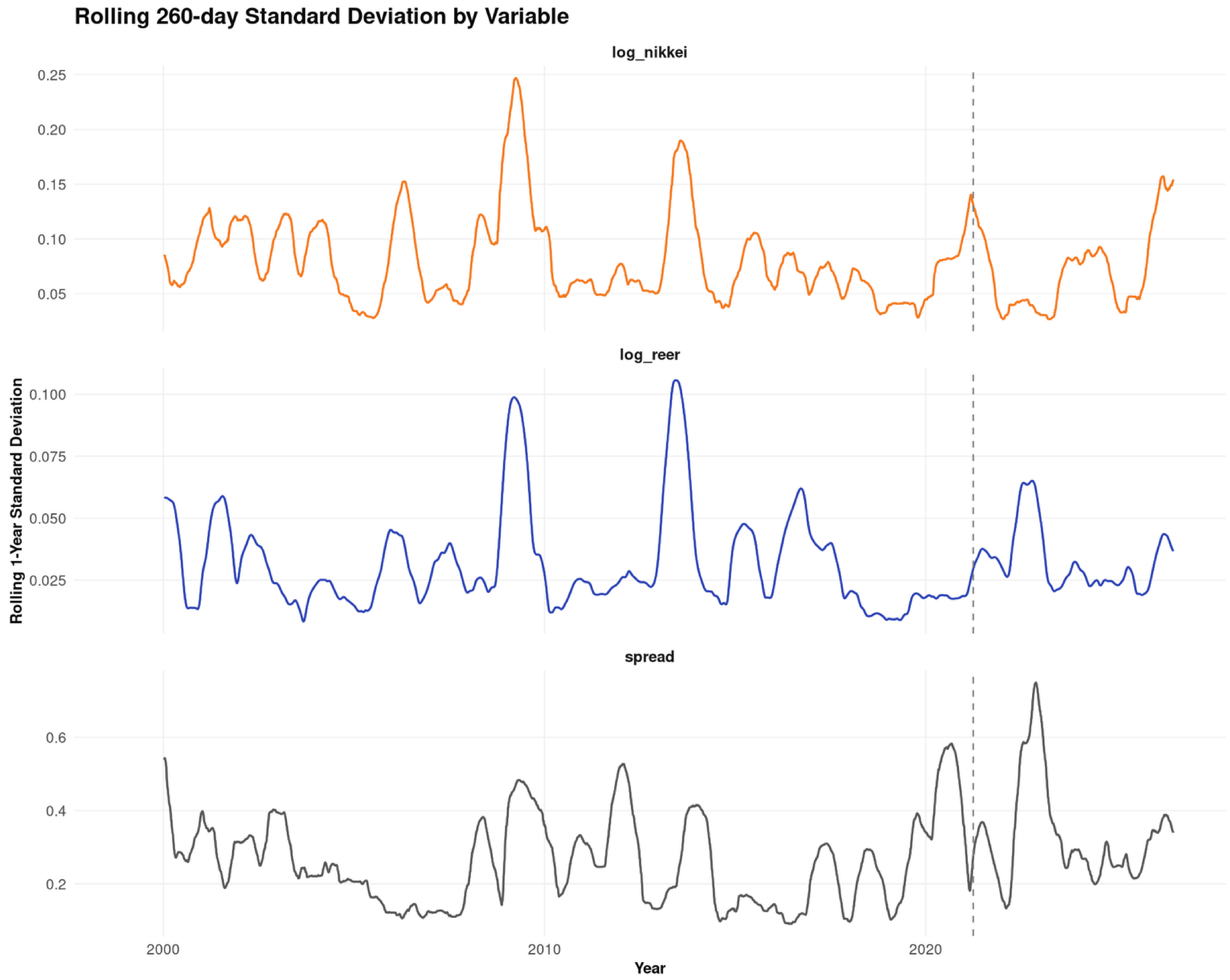
- Sample: 7,164 trading days (5,795 paper / 1,369 extended) after data alignment
- The Risk-off (VIX-based binary indicator) mean fell from 3.2% to 1.9% of trading days. Fewer systemic crises in the extended period.
  - Its standard deviation compressed from 0.176 to 0.137, consistent with shorter VIX spike durations.
- The Spread (JGB 10Y - US Treasury 10Y) mean widened from -2.44 to -2.70 as the Fed hiked while the BOJ remained pinned to YCC.
  - The spread was policy-capped, not free-floating.
- The Log Nikkei (log of Nikkei-225) mean rose from 9.56 to 10.46 (roughly 2.5x in levels) while its standard deviation dropped from 0.33 to 0.24.
  - A post-2021 bull market with fewer crash episodes.
- The Log REER (log real effective exchange rate) mean collapsed from 4.52 to 4.07 (~36% real depreciation in levels) with the standard deviation halving from 0.171 to 0.091.
  - A one-way depreciation rather than the two-sided volatility of the paper period.
- The Log RGDP (log real GDP) mean edged up from 13.19 to 13.28, but its standard deviation collapsed from 0.051 to 0.009 ($F = 31.14$).
  - The paper period contained the GFC, 2011 earthquake, and COVID. The extended period is a flat post-pandemic recovery.
- The Log WUI (log World Uncertainty Index for Japan) mean became slightly more negative (-1.80 to -1.98) while its standard deviation nearly doubled from 0.59 to 1.08. Uncertainty became more bimodal.
  - Sharper spikes, deeper troughs.
- The Debtsec %GDP (net portfolio debt inflows) mean halved from 0.52% to 0.18% of GDP.
  - The only variable where the extended period is more volatile ($F = 0.50$, $p < .0001$).
- The Equity %GDP (net portfolio equity inflows) means are nearly identical (0.22 to 0.17).
  - Stable equity flows across regimes.
- The Other %GDP (net other investment inflows) mean more than doubled (0.39 to 1.13).
  - Driven by yen carry-trade expansion.
- The Direct %GDP (net direct investment inflows) mean and SD more than doubled.
  - Consistent with continued outward FDI.

F Statistic

F-Test for Equality of Variances (1999-2021 vs 2021-2026)

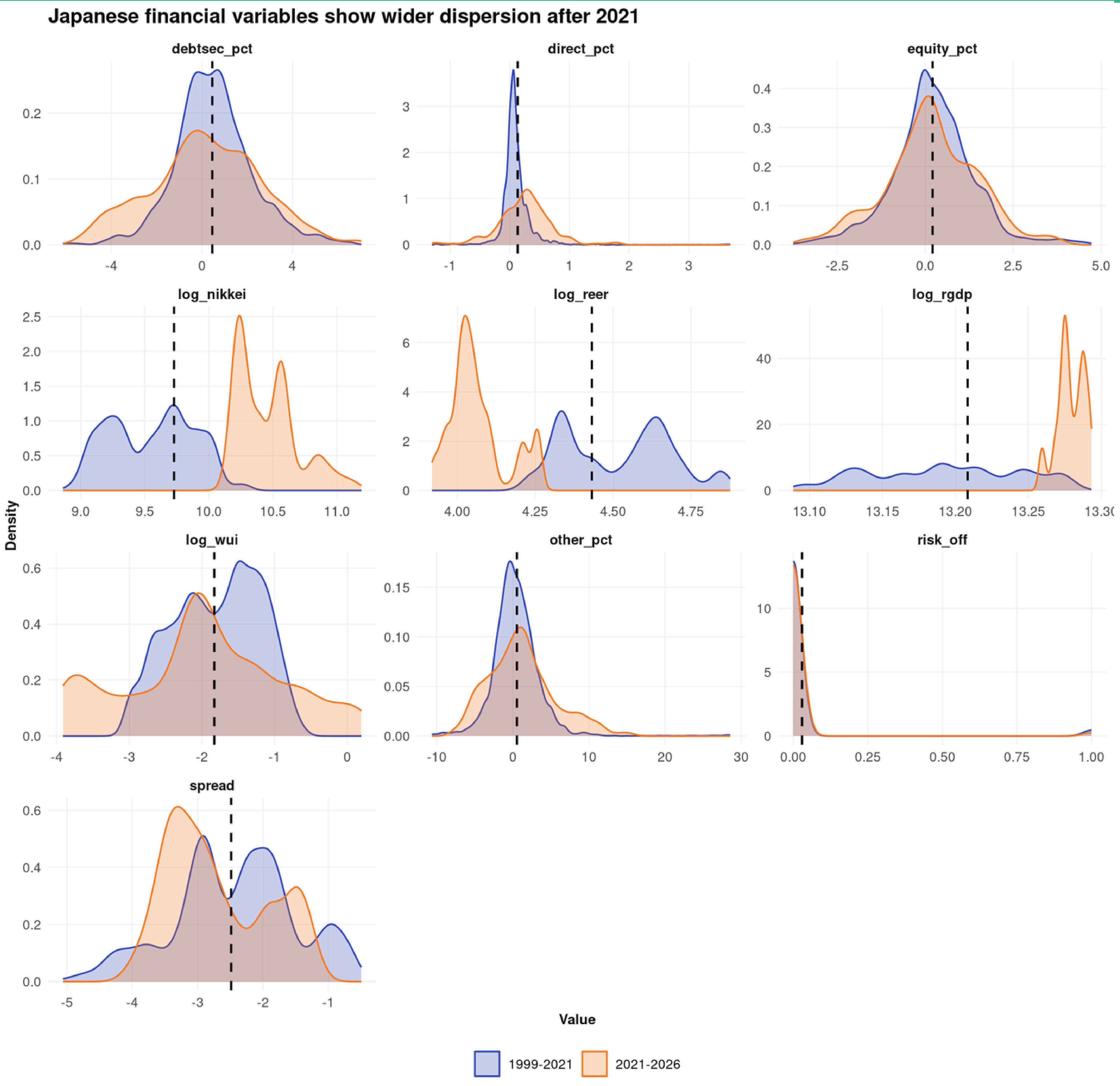
variable	F_statistic	p_value	significant_5pct	higher_variance
log_reer	3.5201	< 0.0001	Yes	Paper Period
log_nikkei	1.8425	< 0.0001	Yes	Paper Period
spread	1.4481	< 0.0001	Yes	Paper Period
debtsec_pct	0.5006	< 0.0001	Yes	Extended Period
log_rgdp	31.1378	< 0.0001	Yes	Paper Period

- The paper period had 3.5x the REER variance. The yen swung both ways during 1999-2021. After 2021 it was a one-way depreciation.
- The paper period had 1.84x the equity variance. Crashes and recoveries in the paper period vs a steady bull run post-2021.
- The paper period had 1.45x the spread variance. The spread was more volatile when the BOJ was not capping yields.
- The extended period had 2x the variance of the paper period (1 / 0.50 = 2). Debt flows became more volatile after 2021, not less.
- The paper period had 31x the GDP variance of the extended period.
    - This contains the GFC, 2011 earthquake, and COVID.
    - The extended period is a flat recovery.



- All three series exhibit significant temporal heteroskedasticity (volatility clustering).
- Log_nikkei Shows high volatility spikes corresponding to the 2008 and 2013-2016.
- Post-2021, underlying base volatility appears lower than the historical extended period.
- Log_reer has massive spikes in 2008 and 2015, but from roughly 2021 onward, its 260-day standard deviation compressed
- Spread shows a drastic shrinkage in rolling volatility after 2021 compared to the high, erratic variations seen in the 2000–2005 and 2010–2020 periods.

- While the title states "wider dispersion after 2021", a statistical inspection reveals this is variable-dependent.
- log_rgdp, log_nikkei, and log_reer display a dramatic narrowing of distribution width
- For log_rgdp, the orange curve is an extremely tall, leptokurtic spike centered around 13.27, indicating massively reduced variance.
- Log_nikkei shows a pronounced rightward shift (higher mean value) with a narrower, potentially bi-modal distribution in the recent period.
- log_reer exhibits a sharp leftward shift (depreciation/devaluation).
- The variables log_wui and debtsec_pct show marked platykurtosis
  - Confirms overall capital flow variables have become more widely dispersed in the later period.



Dashed line: full-sample mean. Variables defined in the reference table above.

- log_nikkei and log_reer are clearly multimodal, reflecting distinct historical currency and equity regimes.
- log_rgdp is highly non-normal (likely bimodal), driven by the structural economic shock and subsequent growth.
- The spread variable shows a heavy left tail
- Direct_pct and other_pct are leptokurtic with a massive spike near zero
- risk_off is a binary state variable, rendered as an exponential-looking decay from zero.
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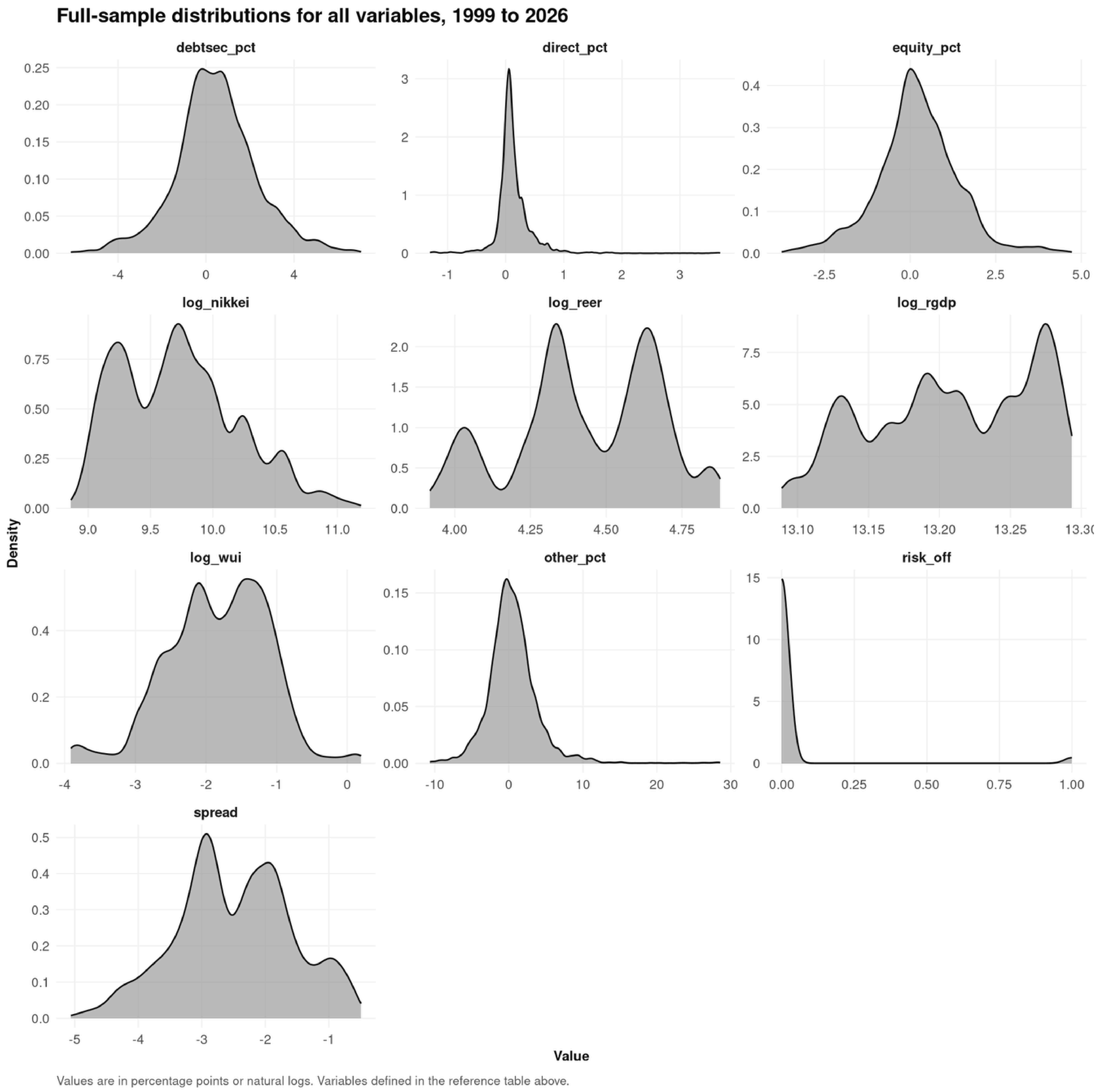
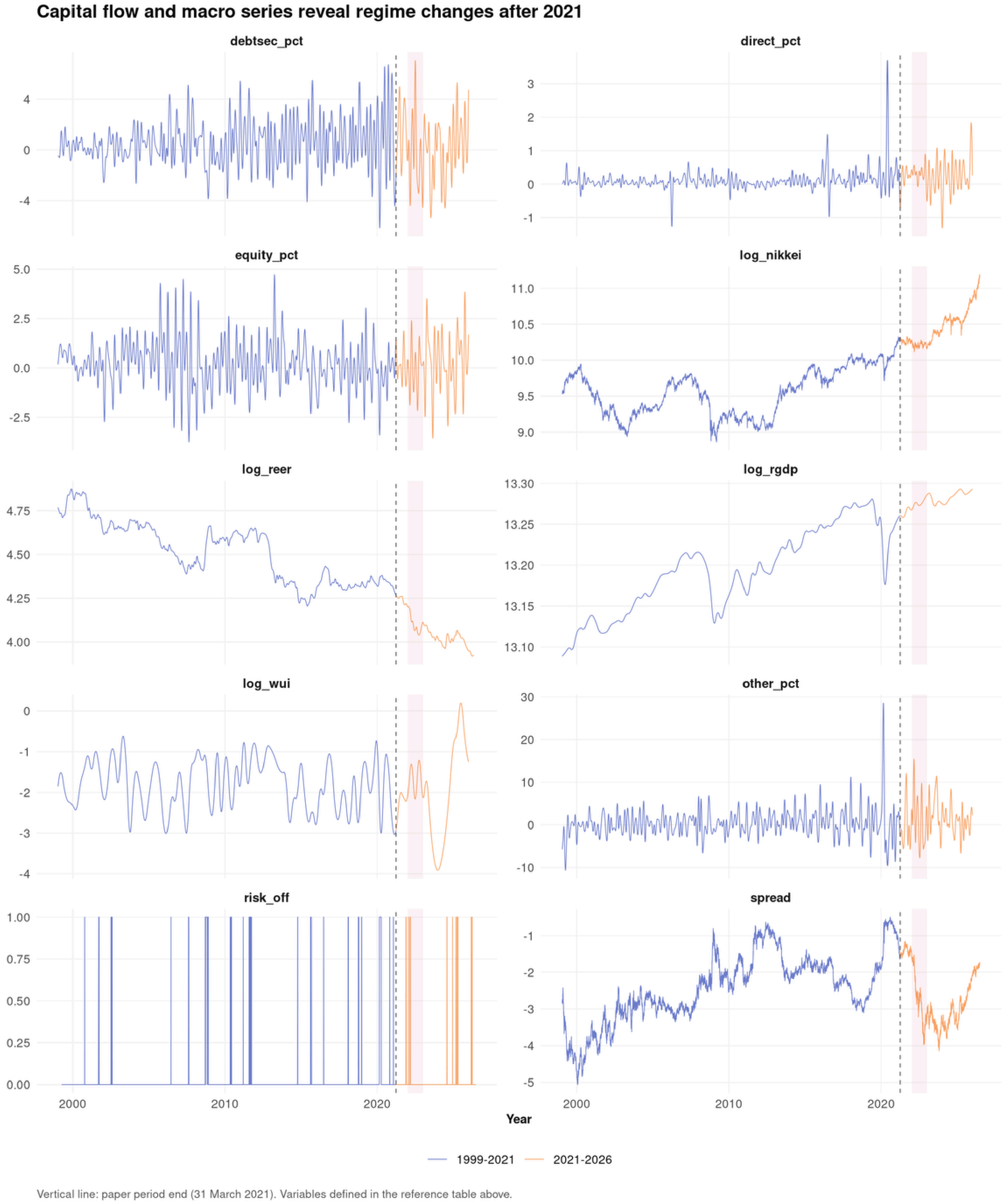
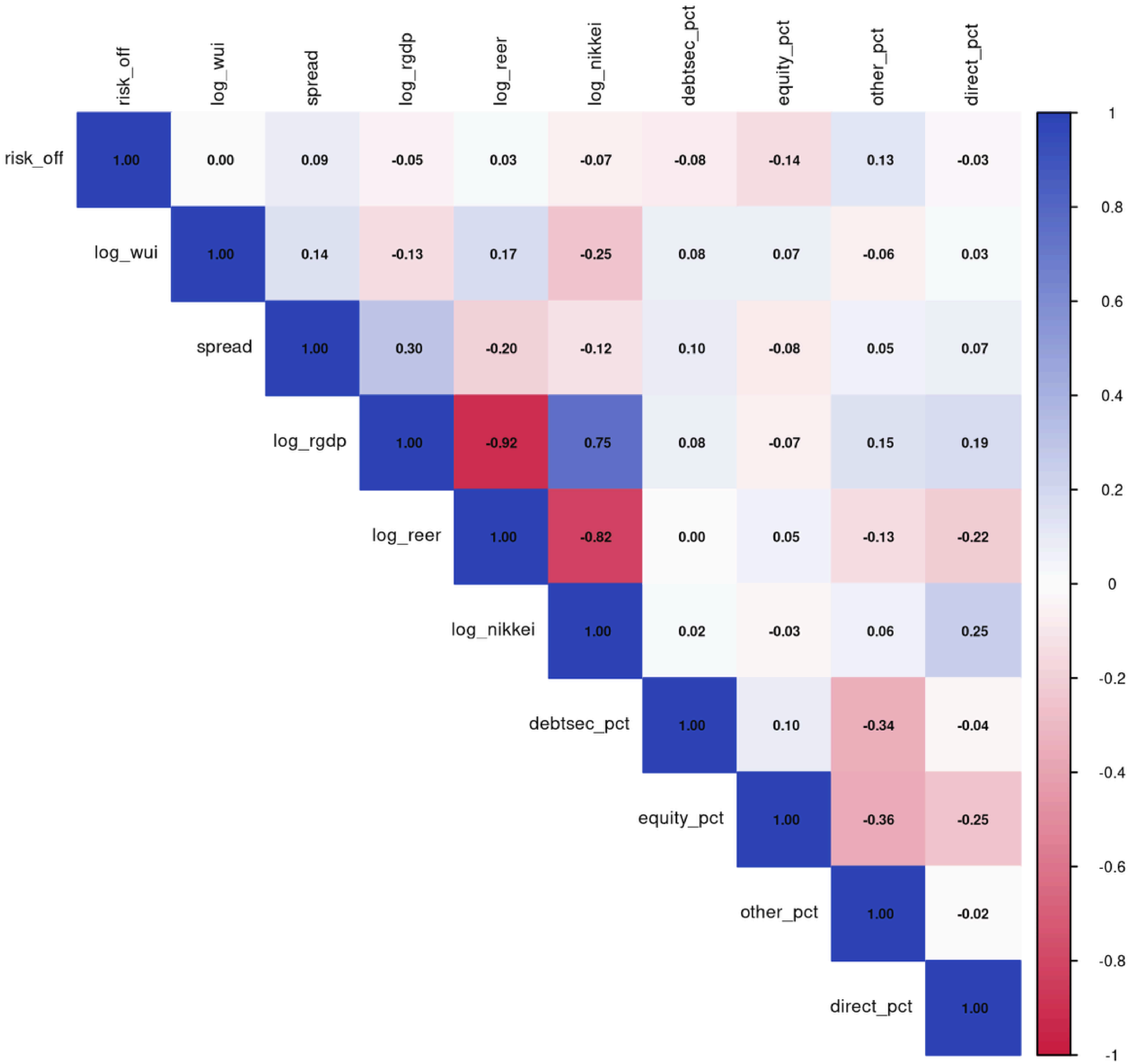




- A definitive regime shift occurs at the vertical line (approximately March 31, 2021).
- log_nikkei shows a steep and persistent positive drift post-2021 that breaks from its prior historical trend.
- log_reer shows a sharp, nearly vertical drop at the break point, settling onto a lower, relatively flat trendline, highlighting a stark structural break in Japanese competitiveness.
- spread establishes a new, lower equilibrium (more negative values) after 2021.
- log_rgdp shows a sharp COVID-19 drop followed by a steep recovery, but the post-2021 is relatively stable compared to the baseline.
- debtsec_pct, equity_pct, and other_pct exhibit significantly increased high-frequency oscillations after 2021, reinforcing the distributional widening observed in the 999–2021 vs. 2021–2026 figure density.

Correlation matrix for the full sample, 1999 to 2026



- Real GDP, Real Effective Exchange Rate, and the Nikkei 225 suffer from extreme multicollinearity.
- Economic growth shares a -0.92 correlation with the Real Effective Exchange Rate, demonstrating that growth is almost perfectly associated with a weaker yen.
- The Yen and Nikkei share a -0.82 correlation, proving that a depreciated yen strongly drives equity market valuations.
- The risk-off binary variable exhibits near-zero correlation with the macro series, verifying that global panic shocks act as independent and idiosyncratic tail-events rather than moving linearly with the domestic baseline.
- Regressing GDP, REER, and the Nikkei against each other simultaneously will trigger severe multicollinearity

NEXT STEPS

- Run the formal Structural Vector Autoregression (SVAR) estimations in R (or python), determine optimal lag lengths via AIC/BIC, and generate Impulse Response Functions (IRFs) to track shock propagation.
- Execute Granger causality tests and compute Forecast Error Variance Decomposition (FEVD) to isolate how much of the Yen's movement is driven by external risk-off shocks versus domestic fundamentals.
- Integrate new theoretical frameworks into the literature review specifically addressing the 2022–2026 Yen carry trade unwind and the Bank of Japan's extreme monetary divergence.
- Draft the results section with a focus on comparing the 1999–2021 baseline against the post-2021 structural break to test the modern safe-haven hypothesis.
- Compile the entire R-based data pipeline and statistical outputs via Quarto into a fully reproducible PDF appendix for the final August submission.

The End

THANK YOU FOR LISTENING

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